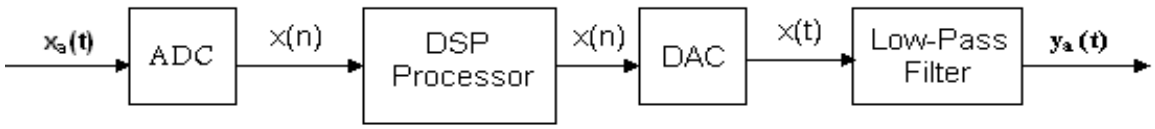
DSP- Revision Sheet # 1 (CH- 1 and CH- 2) – Summer 26

Chapter # 1

1- Explain with drawing the different classifications of the signal.
2- Explain the basic three operations performed in A/D converters.
3- Find the relation between the frequency of the discrete time signal (F), the frequency of the continuous time signal (f) and the sampling frequency (F_S)
4- In detail, what are the main characteristics of the continuous time sinusoidal signals?
5- In detail, what are the main characteristics of the discrete time sinusoidal signals?
6- In detail, Explain the quantization process used in A/D converter.
7- In detail, Explain the sampling process used in A/D converter.
8- In sampling process, derive the sampling condition, Fs ≥ 2fmax.
9- For the following analog signal $X_a(t) = 10 \cos(1500 \pi t)$
   a- Check the periodicity of this signal, If it is periodic, find its fundamental period.
   b- If it is sampled with F_s = 3000 sample/sec, is it possible to reconstruct this signal back from its samples successfully? Find the Nyquist frequency, F_N, for this signal.
   c- Using F_S, Calculate the samples values during one cycle of this signal and then sketch both analog and discrete time signals on one graph.
10- For the following signal: $x_a(t) = A \cos(2\pi f_0 t)$, where, A=2 volt and f_0 =10 KHz.
   a- Check the periodicity of this signal and find the fundamental period.
   b- If this signal is sampled at F_s= 5 KHz, is it possible to reconstruct the original signal from its samples?
   c- What is the minimum sampling frequency in this case?
   d- Using the minimum sampling frequency obtained par (c), calculate the sample values for the first four samples.
11- For the following analog signal $X(n) = 2 \cos(n \pi /4)$
   a- Check the periodicity of this signal and find the fundamental period.
   b- Calculate the sample values for the first four samples.
12- Consider a simple signal processing system as shown in the following figure. The sampling periods of the A/D and D/A converters are T1=5 ms and T2=1 ms respectively. Determine the output ya(t) if the input is given by:

xa (t) = 3 cos (100πt) + 2 sin (250πt) (t in second)



13- Check (✓) for the right statements and (X) for the wrong statements and correct the mistakes.
   1- In quantization process, when the quantization step (Δ) increases, the accuracy of the quantizer increases.( )
   2- Nyquist frequency (F_N), is the minimum sampling frequency ( )
   3- Fourier transform (FT) is calculated for a non-periodic continuous – time signals. ( )
   5- To reconstruct the continuous time signal, x (t), from its discrete time sequence, X(n), the sampling rate, F_S, must be less than the frequency of x(t) signal, f_0. ( )

Chapter # 2

1- Prove that the DFT, X (k), of a sequence x (n) is periodic.
2- For the following sequence, x (n): x (n) = {1, 1, 1, 0}:
   a- Using the direct method (DFT) find the frequency components of the sequence, X (k) and check your results by calculating, x(n), using IDFT method.
   b- Repeat part-a using the FFT algorithm and check your results by calculating its IFFT.
   c- Repeat part-a using the DFT method and check your results by calculating its IFFT algorithm.
   d- Using your results, verify Parseval's theorem.
   e- Calculate and plot the amplitude and the phase spectrums.
   f- What is the computational saving of the FFT algorithm compared to DFT algorithm?

3- For the following sequence, x (n): x (n) = {1, 1, 0, 1}, repeat what you did in problem No.2.